EXECUTIVE SUMMARY

Background
Recent widespread adoption of Generative AI (GAI) (particularly with the public release of OpenAI's ChatGPT (Chat Generative Pre-trained Transformers)) has revealed the potential transformative impact of this technology. It is imperative that AETC explore adoption of GAI technologies to drive improved outcomes or greater efficiency in the Air Force education and training enterprise. AETC A9 and A5 have started a multi-mission collaboration to better understand GAI's potential future place in AETC and the Airman educational environment.

Objective
Begin development of high-impact use cases of Generative Artificial Intelligence in the context of AETC operations.

Conclusions
GAI can potentially improve outcomes in AETC if used to address the six most prevalent career-field skill sets and four high-impact enterprise challenges we have identified. It is likely the most capable technology at present to accomplish this. In response, we ideate nine different GAI use cases that AETC can develop and utilize technology to address. Certain miltary-specific technologies and permissions to use them exist but are often revoked too rapidly to make meaningful impact with them. If AETC is to implement GAI-based solutions, we need to address our gaps in education/understanding, technology/ data infrastructure, and policy.

We also conclude that developing GAI education will enable AETC to compete against near-peer adversaries. Many of China's provinces mandate AI education in primary and secondary schools, so we predict future PLAAF Airmen will be readier than USAF Airmen to utilize the tech upon Force entry. GAI literacy education and AETC-sponsored training may close this gap.

For AETC to compete, we recommend adopting policy to close the GAI education gap. Airmen can then develop and apply the tech to AETC challenges and opportunities.
BLUF: begin development of high-Impact use cases of Generative Artificial Intelligence in the context of AETC operations and recommend strategic competition-based AETC GAI adoption.

Motivation:

Although the basis for Generative Artificial Intelligence (GAI) traces its roots back to the 1950s and 1960s[1], only within the last few years have we seen technologies as effective as today's chat-based AI platforms developed and adopted in the public. Recent widespread adoption of GAI (particularly with the public release of OpenAI's ChatGPT (Chat Generative Pre-trained Transformers)) has revealed the potential transformative impact of this technology. It is imperative that AETC explore adoption of AI

technologies to drive improved outcomes or greater efficiency in the Air Force education and training enterprise.

We start with an overview of GAI, its current role in the classroom, and pre-implementation considerations for AETC. Our ideated use cases along with our methodology to determine AETC-specific technology opportunities follow. We conclude with an overview of how the US and China compare in both GAI education and military applications. Our recommendation is for AETC to initiate a GAI literacy education plan for Airmen.

Generative Artificial Intelligence

GAI encompasses algorithm-based models[2 3] that respond to requests by crafting original content or composing training content based on a statistical training on a large body of data. Most GAI technologies can generally be described in two major components:

1. A model consisting of (a) an algorithm that composes responses by identifying the most statistically likely outputs from (b) large amounts of connected training data (a neural network).
2. An interface where users send queries and receive the responding model outputs.

GAI input and output data currently include text (as Large Language Models (LLMs), with ChatGPT as a leading LLM), images, video, and audio. While non-experts often attribute GAI solely to chatbots and LLMs, different generative technologies can produce music, film, and 2-D/3-D visuals. These models will typically feed its own output data back into its training process, "learning" (and *ideally* improving) over time.

Current limitations of these technologies include numerical calculations[4], prediction[5], systemic bias[6 7], and generally tasks where there is a verifiable right or wrong response. These limitations introduce significant ethical concerns if not addressed throughout the technology's development and deployment life cycle. Despite these limitations, significant value could be captured through the manipulation and generation of text responses of a similar quality to human-written text (e.g., prose or computer code) at high volume with a GAI tool. **Collaboration and constant human oversight is necessary to address these limitations and effectively apply the technology.**

The term "Artificial Intelligence" covers a broad range of technologies, yet not all of them "generate" content. Different non-generative AI technologies may have excellent computational, decision-making, learning, and pattern-recognizing capabilities. For this report, we will focus specifically on generative technologies.

Implementation in the Classroom

Education research centers such as Cornell's Center for Teaching Innovation[8] are considering GAI implementation, learning threats, and opportunities. The advice here and elsewhere varies by discipline, difficulty level, and learning objective. Experts agree, however, that educators must use a nuanced approach to adopting GAI in the classroom, and guidance must evolve as the tools and understanding of effective applications evolve. This and other reports conclude that faculty and administrators need to be in constant communication with each other on the implementation policy.

Additional AETC Considerations

For AETC to responsibly adopt this technology, we need to develop an organization-wide understanding of it. This requires us to address the GAI education gap, implementation challenges, and uses suited for AETC.

Educational Gap

Currently, AETC does not have the personnel to fully understand this technology. Without domain expertise and hands-on experience, AETC cannot pave a path for GAI implementation. We can break down the knowledge gap into:

1. Fundamentals: Maintain up-to-date understanding of what GAI is, how it works, and limitations
2. Technical: Understand hardware, software requirements for use case implementation
3. Social Science: Identify technology's impact vs internal ethnological data and successfully implement safeguards to combat disparate outcomes
4. Security: Build and implement the technology in a way that protects training data, PII, CUI and Classified information
5. Strategic Competition: Close US-China military GAI capabilities gap via education.

AETC action officers can then leverage this knowledge to ensure the most effective use of GAI in their missions and across the organization.

Implementation Challenges

A host of logistical concerns also need to be addressed before implementation:

Gathering data and Data Training: for most real-world and niche applications, experts advise that a reliable model will use at least ~1000 pieces of information. Most industry models operate on billions and trillions. Additionally, creating the framework and relationships between different data entries is foundational to a top-performing GAI technology. Becoming a client to a company such as Scale AI, which currently does collect and curate data for government organizations, may be a promising avenue.

Computational power: AETC can maintain quality by scaling the electricity, processors, storage space, and/or water down to the size of need.

Developing DoD-secure solutions: virtually no pre-existing technology or platform can support training, querying, nor generating info that is CUI and above. Unless a future, DoD-approved company forms, AETC would have to develop GAI tech internally.

Policy: For successful GAI adoption in AETC, we will need to invest time, experts, and a budget that we have not yet allocated to support these technological considerations. Before any of those steps, however, **both DAF and the DoD must adopt clear, consistent, and written policy communicated across the organization that allows AETC to use the tech for the use cases ideated and outlined below.**

Validated Use Cases

Before this technology becomes a part of AETC missions, we must first decide what we will do with it. To provide some early insight, we have developed use cases below that AETC leadership may decide to

validate and resource. We start with a preview of implementation, matching core tasks of example AETC career fields with some of our ideated technologies. Then, we discuss the methodology behind deciding our full list of technology. Following our methodology is the comprehensive list of technologies with detailed descriptions for each. Here, we outline how each technology best prepares AETC to address our most critical challenges.

Use Cases for AETC

In this section, we develop potential use cases for AETC opportunities. We start by reviewing our methodology for ideating use cases and the results of that analysis. Then, we preview how different GAI technologies can intervene in example career fields. Finally, we detail the use cases.

Methodology

To ideate potential AETC use cases, we matched current and projected GAI capabilities, limitations and considerations with select Career Field (CF) expectations as well as larger AF or learner opportunities. Finalizing the list involved:

1. Analyzing CFETP data for expectations of high developmental potential:
    a. Examining the descriptions contained in each of the ~150 unique CFETP documents[9] spanning over 130 unique career fields found on the AF's e-pubs website. We used the language found in the roles and responsibilities (for >90% of the Career Fields). If no Roles and Responsibilities section was found, we used language contained in the Knowledge section (~10%) or, if also not found, then in Competencies (<1%).
    b. Identifying shared traits between CFs (single keywords and key phrases) with natural language processing (NLP)-based frequency analysis in Python programming language
    c. Focusing on shared traits found in >10% of CFs.
2. Ideating how different generative technologies can create solutions that fit:
    a. these traits, skills and needs from CFETP description data.
    b. Additional applications added for learners or USAF opportunities in general, which were identified by examining both AF-related and student-related news.

Results of CFETP documents and Analysis of USAF opportunities

Below is a summary of most-cited tasks in AETC career fields' responsibilities and knowledge requirements. The keywords/phrases, total CFETP count, and % of total CFETP descriptions that contain the respective key phrases are outlined in the table. If AETC increases success at these tasks by adopting GAI, we predict that these areas potentially have high Impact.

High Impact areas from Career Field descriptions:

1. Equipment (operation, maintenance): 100: 68.03%
2. Planning/coordinating: 93: 63.27%
3. Management (process/plan/procedure/program/talent): 89: 60.54%
4. Data (analysis/management/storage): 74: 50.34%
5. Safety (weapons, hazards, aid): 66: 44.90%
6. Medical: 19: 12.93%

Our additional analysis of AF news and internal Impact opportunities identified urgent challenges:

7. Learner-centric education implementation[10]
8. Recruitment shortages[11] [12]
9. Fitness levels[13] and mental health[14] [15]
10. AF casualties [16], hazards[17], and inflated costs[18] in training exercises.

Example Career Fields

Out of the over 130 Air Force Career Fields, generative AI can potentially improve Airmen's abilities to fulfill the roles and responsibilities for virtually each one. Select examples in the field may include:

9S100: Scientific Applications Specialists may find a **GAI data analytics hub** useful for their analysis tasks, utilize a **tutor bot** to become proficient in or refresh their foundational chemistry, physics, and math, and utilize a **GAI training and expertise technology** to assist in their on-the-field engineering and other technical knowledge tasks.

1C8X3: Radar, Airfield, and Weather Systems (RAWS) maintenance personnel can utilize a **GAI data analysis** hub to interpret technical data and blueprints, brush up on using mathematical principles to solve electronic formulas via a **tutor bot**, and work with the guidance of a **training and expertise technology** to have real-time assistance in analog and digital electronic circuits, maintenance data processing systems, Air Force maintenance and supply procedures, and basic soldering techniques.

Applications of GAI in AETC

Here we provide a mapping of the Impact area to the ideated technologies. Each ideated technology addresses at least one identified opportunity, and each opportunity applies to at least one technology.

Ideated Technology	Impact Area(s)
Tutor Bot	7
Data Analysis Hub	4
Health and Fitness	9
Talent Management Assistant	1, 2, 3
Recruiter Bot	2, 3, 8
Teaching Assistant	4, 7
Supervisor Assistant	4, 7
Front Office Management Assistant	2, 3, 8
Trainer and On-the-Job Guide	4, 5, 6, 10

We also provide detailed descriptions for what these ideated GAI technologies might look like in AETC. For each of these applications, we include—Function: Primary technology purpose. Problem: Core opportunity the tech addresses/Impact opportunity. Requirements: Operational components of the

proposed tech. Why: Reason that this tech is the solution as opposed to other options. Examples: Implementation in AETC; Challenges: Projected difficulties with implementing this tech. Future: Further development and advanced versions of the proposed tech.

The following list is a snapshot in time, reflecting the known current state of GAI as well as current unmet needs in AETC. It can vary greatly over time and by enlisting a wider array of expert opinions.

Tutor Bot – Potential Big Swing

Function: LLM-powered chatbot that guides student learning through text-based interaction.

Problem: The Air Force's transition to the learner-centric educational environment has been hindered by resource shortages, such as instructors and knowledge, experience of learner-centric teaching approaches, and technologies that guide learning rather than regurgitate (correct or incorrect) answers.

Requirements: An LLM model with a chatbot interface; subject-specific and tutor/guidance-style training data.

Why: Benefits to adopting such technology is twofold:

1. Making student-centric learning possible: a growing body of research from academia[19] and learning research centers[20] strongly suggests that individualizing the learning experience to the student improves students' ability to learn. Using an LLM-powered chatbot reduces the manpower cost in instructors to implement and support student-centered learning.
2. Automated guided learning: Unlike the style of chatbots currently in the public (designed to most efficiently deliver the most accurate responses to user prompts), this version of the technology guides students through the learning process.

Examples:

a. Enhancing digital classes: Air Force e-learning, Air Force Digital University, or DAU courses/books/materials can come accompanied with this AI tutor bot plugin so that learning is enhanced at pace with the student's progress in their course.
b. In-person and homework skills building: when students have a topic-related question either during their class or when completing independent assignments, they can ask their chat bot tutor for concept explanations or problem-solving guidance.

Challenges: The training data needs to be updated for evolving subjects or comprehensive for well-established ones. Additionally, the model needs to be trained to guide foundational skills-building rather than produce the answer right away. For CUI and above, ***developing a solution would need to be funded by and grown within the DoD.***

Future: As this technology advances, it should be able to gauge an individual student's learning style and adapt its instruction method according to student responses. A multi-modal version of this tech—through video and augmented reality (AR), for instance—can supplement text-based learning.

Data Analysis Hub

Function**:** A prompt-based suite of visualization, calculation, and text-based generative technology that offers analysis insights, processing (modeling, synthesis on real data capabilities), and problem comprehension assistance.

Problem: AETC is experiencing a personnel shortage to complete data science and analysis tasks. Underperforming in these areas undermines AETC's capability to utilize data in decision-making processes and day-to-day operations.

Requirements: This technology would consist of a data lake, visualization tools, calculation capabilities, and a GAI input interface for both trained analysts and non-trained personnel to better understand their data-centric problems. The interface can be used to input visual or text-based information to process. It can also be used to "talk" through the problem itself, ways to approach it, and interpretation of results.

Why: Whether mastering fundamental analysis skills or crafting and deciding a best analysis approach, personnel of all experience levels thrive on collaboration to understand and interpret data-motivated, evidence-based problems. Teams often expend time teaching skills in real time and/or discussing a range of approaches that may or may not work. If teams can instead use a tool that (a) cuts down on skills learning time, (b) consolidates expertise, (c) decides an analysis approach, and (d) interprets results, then AETC can save time on high-quality deliverables.

Examples:

a. Envision plugin: much of AETC data used in customer-facing reports comes from Envision and needs to be manually compiled and query-based or manually interpreted. An AI database plugin can cut down time on gathering, standardizing, and prepping the data for analysis.
b. User API: Airmen can also have this application accessible to them on their local devices, ready for use from every stage of the data project: interpretation of problem, brainstorming approaches, data entry, analysis, visualization of results, interpretation of results, and customized presentation of results depending on customer.

Challenges: besides the standard CUI-handing concerns, data-specific hurdles may include

a. Computation: while LLMs are currently weak in calculations, connecting a tool such as Wolfram Alpha would be an option.
b. Data centralization: AETC needs to consolidate and standardize data so that personnel are more likely to get accurate results for their projects.

Future: With missions that require collaboration or real-time data understanding, this technology should be hosted on a platform that integrates with fielded technologies and multiple users at once.

Health and Fitness

Function: provide personalized health and fitness guidance to Airmen through text-based and visual-based queries.

Problem: AF personnel and new recruits alike face increasing challenges in meeting and maintaining fitness standards. Failure to meet these standards may disqualify them from service, or they may receive penalties that jeopardize their professional standing. Additionally, Airmen face similar personal and professional risks when they feel unable to seek out necessary mental health help. This has resulted in an uptick in on-base violence and suicide rates.

Requirements: an LLM model with a chatbot interface.

Why: addressing the problem with GAI is threefold:

a. Manpower: USAF likely does not have enough personnel to establish fitness camps (like the ones that have succeeded in small groups) to train future Airmen. GAI can step in when a real person cannot.
b. Long-term fitness companion: Without this tech, each AF military member is solely responsible for maintaining individual fitness levels throughout their careers. GAI can personalize health plans to each Airman's needs and capabilities, especially in addressing the increasing prevalence of weight disorders.
c. Accessibility: Increased digitization of mental health resources in civilian life has made seeking help more affordable, efficient, and accessible for many. Airmen may find it less intimidating to interact with GAI for mental health support than from a person. This can encourage Airmen to be more candid, allowing USAF can safeguard their wellbeing.

Examples: AETC can use this technology as:

a. Personalized fitness goals and planning: the GAI can use an Airman's physical stats, current eating/exercise habits, and fitness goals to deliver a personalized plan for success.
b. Therapist: The technology can serve as a therapy bot, providing mental health support and check-ins for Airmen, either for addressing existing challenges or for proactive mental health maintenance.

Challenges: If AF personnel cannot oversee all the interactions between Airmen and this technology in real time, safe and responsible interaction is imperative. The tech needs to undergo rigorous safety testing and Airnen need to undergo responsible use training. Implementing safeguards into the model may also be necessary to avoid misuse and dangerous outputs.

Future: The model can access real-time data from smart technologies or internal personnel databases and therefore generate real-time advice for Airmen.

Talent Management – potential big swing for TMOs

Function: Chatbot offering career advice to Airmen and/or placement advice to Talent Management Officers.

Problem: When Airmen consider their career paths, they may be unsure of where they want to go or what they need to do to get there. TM offices, if faced with a candidate pool too small or too large for must-fill positions, may also not be able to efficiently decide placements.

Requirements: an LLM model with chatbot interface

Why: A GAI model can use force-wide aggregate data to help both TMOs and Airmen efficiently make informed decisions about career placement. The LLM-based chatbot interaction allows both parties to address their questions and discover placement opportunities based on a user's individual questions and concerns.

Examples: Both Airmen and TMOs can use this talent management assistant to optimize their career path discovery and placement:

a. Airmen can interact with an LLM-based chatbot, providing the bot with their own aspirations, career history, and concerns. The bot would deliver recommendations based on historical career trajectory data and this input.

b. For saturated candidate pools, TMOs can provide current candidate pool and must-fill position information. The model would then use this info to provide a few rank-ordered options for filling the position.
c. For personnel shortages, TMOs can provide descriptions of their must-fill positions, and the bot can deliver options of missions/organizations to source candidates.

Challenges: **Ethical concerns with perpetuating existing biases would need to be addressed with intentional safeguards that help promote diverse and representative leadership.**

Future: For TMOs, the model can automatically access internal personnel data and recommend placements. For Airmen, the model can access their personnel profile and customize a career trajectory and/or course of action.

Recruiting– Potential Big Swing

Function: LLM-based chatbot that interacts with civilians about AETC/AF opportunities and onboarding.

Problem: The entire military faces a critical personnel shortage, posing a national security risk.

Requirements: The chat bot would interact with potential recruits like a human recruiter. It must be able to accurately inform a user about AF opportunities, the onboarding process, and any other information a recruiter typically covers. Ideally, at the end of the interaction, the bot would connect a user with a human recruiter, using geographic data and conversational recall.

Why: Other recruiting incentives thus far have not successfully filled in the personnel gap. Deploying a relatively inexpensive and personable option *may* increase enlistment.

Examples: Add chatbot feature to our websites with highest civilian traffic.

Challenges: Besides gathering comprehensive recruiting data to train the chatbot model, misuse of the chatbot can also be of concern. If a user prompt engineers to cause a hallucination or a malfunction, our bot may deliver incorrect data or reveal PII/other confidential data.

Future: A hyper-realized version of this would pool public data from online platforms (e.g. web cookies, selective ad targeting, etc.), posing a significant ethical concern. A more realistic future goal may involve interactive or gamified GAI ads that do not abuse the public's data.

Teaching Assistant– Potential Big Swing

Function: an LLM and visual-based tool that interacts with instructors to generate course content and automate content-based classroom tasks.

Problem: Current staffing shortages limit AETC's ability for instructors to craft curricula that teach necessary skill sets while limiting the potential for plagiarism.

Requirements: This technology would consist of an LLM trained on the taught subject(s), learning stage of students, and the chatbot interface.

Why: Teachers can spend more time interacting with students for a personalized learning experience if they are able to save time on developing/vetting course content and other classroom-related paperwork.

Examples: Teachers can use the LLM to generate syllabi, reading lists, problem sets and classroom example problems, and grade papers.

Challenges: As with the tutor bot, a teaching assistant bot needs to understand both the subject matter and ways that students will most effectively learn the material.

Future: A teacher can use an LLM like this in conjunction with data-keeping capabilities to track student progress. The LLM can use student progress to adjust the curriculum and suggest student-specific instruction techniques. Additionally, the assistant may even guide its own class or course as an instructor.

Supervisor Assistant:

Function: LLM-based chat bot to assist decision-making for front-line supervisor

Problem: Managers can find themselves overwhelmed by the task to create the best teams they can when teams constantly change or face staffing shortages.

Requirements: Chatbot interface, LLM, historical personnel data, career field requirements data.

Why: Whenever experienced leaders may be unavailable to offer advice, or if team data cannot be shared, a well-trained chatbot can efficiently give custom advice to managers/supervisors on their concerns to better manage and structure teams.

Examples: Supervisors can use this LLM-based chatbot to ask about personnel placement, ways to fill current team gaps, advice on earning team trust, and other effective management guidance.

Challenges: Besides the need to centralize data for making the best-informed personnel decisions, the **ethics** of using historical data is concerning. If it is used to project future force development, then biases of the past perpetuate into the new ranks. ***Deciding which intentional safeguards best avoid implicit bias must be at the forefront of this technology to best serve supervisors as well as their teams.***

Future: The LLM model would also be able to interact with real-time personnel data (such as current staff and their performance, interests and career goals, historical career record, and proficiencies) to optimize team building for both managers' and team members' goals. Managers can request data pulls and automated optimal team configurations based on custom factors.

Front Office Management Assistant

Function: LLM-based bot that automates both simple and complex (multi-step and multi-app) clerical tasks.

Problem: Many front-office tasks, when done manually, are tedious, time-consuming, and too abundant for staff to manage. This both introduces a greater likelihood of error as well as time taken away from accomplishing AETC missions.

Requirements: The bot would consist of an LLM to understand user's tasks as well as the ability to "plug in" to other office apps to accomplish the task.

Why: An electronic assistant that can intelligently interpret and execute on these tasks decreases the need for office management to do so as well as increases workflow efficiency.

Examples:

a. Email/chat distribution and customization: AI assistant can help office management decide best ways and times/time routines to send communication and reminders based on importance, subject, recipient, etc. As a plugin, it can also automate the communication itself.
b. Conflict-free scheduling: automatically select most optimal time for members to meet within a specified time frame, send out invitations, and schedule
c. Paperwork: The plugin can interpret the document, determine what info is needed, and where the info and document will eventually go. It can also automate the tasks of locating necessary info, filling it out, and sending it to the appropriate destinations.

Challenges: Besides handling CUI and above, investing in technical expertise to implement this bot may require overcoming software hurdles. The most immediately feasible version of this technology is a chatbot, where a user can enter requests and then manually copy over the content the chatbot produces.

Future: FOM tasks may be integrated in a single system-of-systems (SoS) technology based in GAI. For instance, a user may ask a chatbot to complete a multi-app task, and the chatbot's instructions can inform integrated technologies to automatically complete that task.

Trainer and On-the-Job Guide

Function: Drill essential Airmen skills by simulating occupation-based scenarios.

Problem: The Air Force continues to lose lives and overspend its budget by sending Airmen out undertrained or into hazardous/uncertain environments.

Requirements: An LLM-based chatbot trained on mission-based scenario data can train Airmen on step-by-step scenario-based responses. An alternative option is a Virtual Reality/Augmented Reality (VR/AR) environment powered by visual/audial GAI.

Why: Currently, GAI is the most technologically advanced option that can simulate dynamic, lifelike environments. AETC can scale realistic, minimal-risk training by using GAI.

Examples:

a. LLM-based chatbot that tests Airmen on responses to possible on-the-job scenarios
b. VR/AR multimodal environment where Airmen can interact with their simulated environment

Challenges:

a. Gathering enough reliable training data for hard-to-predict or new scenarios
b. Integrating multiple technologies to create a multi-modal, immersive simulation

Future: While Airmen can greatly gain confidence from training in simulated environments, personnel in many career fields (entry level or otherwise) could greatly benefit from on-the-job assistance out in the field on a real mission. A future version of this technology could be a suite of generative tools that both (a) help the user troubleshoot their on-the-job questions and concerns in real time and (b) train Airmen on common and emergent troubleshooting techniques.

Potential Big Swing

AI Flying Training Instructor – develop a capability (see below)

AI Talent Management - see above for TMO description

Potential future applications for the trainer may include:

Engineers and Maintenance: Real-time aircraft maintenance supervision and assistance: intaking visual scenario and delivering visual and chat-based prompts to guide user's troubleshooting. Additional training and guidance on **safety** practices.

Wargaming in PME: virtual GAI can generate new wargaming scenarios for Airmen to both train critical well-known skills as well as reveal new skills that might be necessary to tackle a new wargame.

Operations Research Analysts: Supply chain optimization in unexpected events: If a national event or global disaster occurred, USAF would still need to know how to obtain its supplies as quickly as possible. Reinforcement Learning (RL) can generate several optimal possibilities on achieving this in a critically dynamic environment.

Coding: **programmers and software developers** can develop custom code and debug. This is already a strong LLM capability.

Cybersecurity/IT: IT specialists can take advantage of GAI pattern recognition in developing internal safety and security tests

Medical: Medical professionals can use data aggregation to establish Force-appropriate health standards and combine this with individual patient information to make informed, personalized healthcare decisions. They may also receive real-time assistance with urgent and difficult decisions. On the learning side, training in a simulated environment can help students gain life-saving skills while reducing mortality risk.

Flight and Pilot Training: Pilots-in-training can enter an immersive flight simulation. GAI can develop visuo-audio scenario testing environments. These environmnents can train pilots how to respond to unexpected hazards before on-field training, reducing mortality risk and equipment cost while increasing training schedule flexibility.

Tech Training: Currently, Airmen can work toward a college degree on one of 22 technical training tracks[21] while progressing in rank to become skilled and specialized in their fields. GAI trained specifically in the knowledge base and technologies that Airmen in the nearly 140130 career fields

Evaluating AETC Priorities for Use Case Adoption and Performance

As previously mentioned, the use cases we have identified here represent a snapshot in time of information and expertise contained within our team and contacts. If leadership or future teams would like to determine their own use cases to pursue, they may either decide to prioritize a number of the ones we mention or create their own. In this section, we provide frameworks for both options. For prioritizing ones found here, we take a Deep Dive into use cases that leadership (at the Col level) has expressed as potential Big Swings (significant enterprise impact). To create new use cases, we outline a framework based on the SPARK method that determines high-impact competencies and opportunities using non-AF data.[22] For analyzing how effective GAI technologies may be in fulfilling use cases ideated with any

method, we outline a modified version of the second part of the SPARK method that tests use case competencies against GAI performance.

A Deep Dive into Potential Big Swing Use Cases

While we have identified use cases above, insights from AETC Col-level leadership help us narrow down the list to core use cases that we recommend AETC prioritize in the near future. We categorize these as potential "Big Swing" use cases that **address at least one *enterprise-wide* opportunity.**

An Alternate Use Case Generating Framework via AETC Competencies Predictions

The list of Impacts we generated (based on Career Fields roles and responsibilities as well as AF-specific news) underly our use cases. Therefore, our claim that these use cases best represent the needs of AETC relies on the assumptions that (a) Impact-based information consists exclusively of AF-related information, and (b) this Impacts information will be relevant long enough to make the adoption of the tech worthwhile. Other resources and future insights, however, may also provide equally or even more useful high-Impact areas. New high-Impact areas could inform more use cases, and so we strongly recommend users routinely analyze which competencies the enterprise will need using both internal and external resources.

One possible method to generate areas of Impact that later inform use cases involves humans interacting with the GAI technology to break down a series of tasks into task-agnostic competencies. Namely, personnel engage in human-machine collaboration to determine a comprehensive list of competencies that are suited specifically to their mission or goals. This not only allows the human user to break down their specific mission into its core components, but it will also give the human user insight into *how* the technology of their choice will think about and approach the task at hand.

For instance, if an Airman is considering using one of five different chatbots for a data analysis task, the Airman can ask each chatbot what it thinks the necessary skill sets or competencies are required to complete the task. Each chatbot, tentatively, will initially produce its own list followed by other suggestions if prompted to do so. The respective lists will give the Airman some insight not only into their own data analysis task but also how each individual chatbot would approach the problem. This would help refine the Airman's use case to determine if the technology can perform for that use case or use cases requiring similar competencies.

Users can also inquire about the (a) weighted importance or influence of each competency or (b) the interconnectivity of the identified competencies. With weighted importance, for instance, a chatbot may say that data analytics comprises 40% computational skills and 60% reasoning, etc., but that accurate computation is important to effectively reason. This offers further insight into the areas in which a GAI-based use case needs to perform well to better the enterprise.

Projecting Potential GAI Use Case Performance

A GAI user in AETC may have areas of High Impact and potential use cases from any of the following methods: extract directly from this report, use the framework we just outlined to come up with their own, or derive Impact and use cases separate from these methods entirely. In any of these circumstances, the user should also be able to determine *how* effective adopting a use case-based GAI is for their particular area of Impact.

Our framework for grading performance is borrowed from the FLASK (Fine-grained Language Model Evaluation based on Alignment Skill Sets) method[23]: If a user has a known task with an established set of competencies, they can create and deploy a task-agnostic competency test. For each competency, the user can generate sample questions of varying difficulty that test that competency. Depending on how many questions the GAI technology answers for each difficulty level and competency, the user can conclude that *at best* the technology can, if adopted as a use case, perform as effectively as their performance score. If a user weights this score by the proportion of influence a particular competency has on a given task, then the user can estimate how effective a use case technology will be on their task.

US vs China

We predict that the upcoming generation of military personnel from our most prominent near-peer adversary—the PLA—will be better prepared than future US military to utilize the full capabilities of GAI upon Force entry. China's emphasis on mandatory GAI curricula and early adoption of US and homegrown GAI technologies have and will continue to put Chinese ahead of US students in this space before they become our recruits. To understand how we can best support Airmen in closing this proficiency gap, we must first understand China's current progress and future vision for educating the next generation in GAI.

Education

Since 2018, China's Ministry of Education mandated AI curricula in its high schools[24]. Many of China's provinces have swiftly transitioned in this mandate by implementing GAI education in primary and secondary schools since 2019, paving the way for more ambitious policy. These policies refer to (1) utilizing GAI tools to enhance classroom lessons and (2) teaching a curriculum on GAI itself (analogous to Information Technology or software development studies, for instance). Besides classroom implementation, China also boasts (3) education-centered, publicly available GAI tools. Here we provide a brief breakdown.

(1) Lesson enhancement: Some Chinese classrooms already feature some of the capabilities in our ideated technologies. Teaching assistant bots offer personal guidance based on each student's individual performance, generate curricula/course content, and hold conversational lessons with students[25]. Instructional use of GAI in the US, however, is institution specific. Education experts offer nuanced, situational, and subject-specific guidance[26].

(2) Curricula: Local and province-wide educational policies have already taken root, affecting millions of Chinese students at primary and secondary schools across the country. In March 2023, Zhejiang[27] announced that AI core curricula would become compulsory in primary and secondary schools[28] with 1,000 experimental institutions by 2025[29]. It would build from the over 700 institutions already constructed up to that point. Cities such as Nanjing[30], Shenzhen, Shanghai and Beijing[31] have launched their own programs as well, with some students building their own applications[32]. Renming University's affiliated high school located in Beijing, for instance, launched its "Artificial Intelligence + X" curriculum, which includes machine learning, deep learning, data mining, automatic driving, natural

language processing, and computer vision. Information Technology and AI are even integrated in many of these cities and provinces' versions of the national standardized test (gaokao).

(3) Educational tools: Chinese students already have access to personal-use educational GAI tools. A number of schools in China have already implemented a technology resembling our proposed tutor bot Squirrel AI[33][34] to help struggling students[35] and to reduce student burden[36]. A GAI (Wenku) on China's most popular browser (Baidu) offers chat assistance and multimedia content generation[37], often used for academic assignments.

Military

Policy: Both the US[38] and China[39] [40] [41] have outlined policy to make AI trustworthy before full-scale implementation. China has gone further, however, to assess its current work force capabilities and strategize for the future. While a lack of concrete policy appears to hinder AETC's ability to use GAI, China promotes AI standardization and application in the military[42].

Technology: As of this report, the US military has assessed a small variety of LLMs and chatbots. Different branches have sanctioned use of some for certain CUI applications. AETC had for a time approved use of ChatGPT+ and AskSage AI (an extensible GAI platform hosting a number of industry-leading LLMs, including GPT 3.5 and 4). In June 2023, the Navy approved a conversational tech support chatbot called Amelia[43]. Other LLMS include NIPRGPT as a general-use chatbot as well as Percipio, resembling a cross between our Supervisor Assistant and our Trainer. Additionally, AETC has recently approved a trial use of Infotech[44], comprising a set of various AI instructor-guided courses resembling the future vision of our Teaching Assistant. As of publishing, we have not identified Chinese military-specific GAI technologies.[45] The Chinese military, however, is also ideating potential use cases and applications, such as for staff assistants. [46].

Careers: Several Chinese military academies list AI-centered careers on their job ads[47]. They range in recruitment, algorithm engineer, and software engineers ranging from recent graduates to experienced professionals. The US military's desire to explore and adopt GAI outpaces our preparedness; as of writing, there are no publicly-listed AI careers in any of the military branches advertised[48]. The Chief Digital and Artificial Intelligence Office (CDAO), however, posts various leadership and engineer roles[49] related to Task Force Lima[50].

Preparedness for (Allied) Collaboration

The US and Chinese preparedness for international collaboration vary by individual policy and peer availability but mutually resemble US-China policy. The White House's Executive Order[51] emphasizes the US's vision to lead in AI development but to collaborate with allies on policy, technology, and mutual interests. A part of China's national strategy is international joint research and development efforts with leading scientists[52] . None specify that these collaborations are restricted to particular countries or even allies, though in 2019 China and Russia signed a cooperation in rising digital technology such as AI[53]. Just as important as our preparedness to collaborate with allies is our allies' preparedness in this space.

Whereas China is the only country in its strategic partnerships with robust national GAI infrastructure[54], an array of US allies may be able to collaborate in this space[55] [56]. As for US-China cooperation, both countries routinely collaborate on technological development[57] and agree to keep communication open on strengthening ethical GAI policy[58].

Conclusion/Recommendation

AETC understanding of Generative AI as well as its potential and consequences vary greatly throughout the enterprise. Lack of clear policy that can endure rapid developments in this transformative space will cause our Airmen to lose their global competitive edge. If we emphasize GAI literacy development, implement secure, ethical safeguards, and consider adopting any of the ideated technologies mentioned before, we may optimize Air Force education and training.

BIBLIOGRAPHY

1 https://matthewdwhite.medium.com/a-brief-history-of-generative-ai-cb1837e67106
2 https://research.ibm.com/blog/what-is-generative-AI
3 https://www.mckinsey.com/featured-insights/mckinsey-explainers/what-is-generative-ai
4 Empirical testing of an array of LLMs through AskSage demonstrates that an answer may depend on the wording of a query, contextual patterns, or other language-based variables.
5 AI Snake Oil by Sayash Kapoor. Accessed 2023. https://www.aisnakeoil.com
6 https://guides.osu.edu/newjimcode/what
7 https://www.nist.gov/news-events/news/2022/03/theres-more-ai-bias-biased-data-nist-report-highlights
8 https://teaching.cornell.edu/sites/default/files/2023-08/Cornell-GenerativeAIForEducation-Report_2.pdf

9 https://www.e-publishing.af.mil/Product-Index/#/?view=search&keyword=CFETP&isObsolete=false&modID=449&tabID=131
10 Examples of student-related news include national declining proficiency scores in math and reading. https://www.nationsreportcard.gov
11https://www.nbcnews.com/news/military/every-branch-us-military-struggling-meet-2022-recruiting-goals-officia-rcna35078
12 https://www.pbs.org/newshour/show/why-recruiting-and-confidence-in-americas-armed-forces-is-so-low-right-now
13 https://www.npr.org/2023/10/05/1203766333/us-army-military-recruit-pentagon-air-force-navy 14 https://www.npr.org/2021/06/24/1009846329/military-suicides-deaths-mental-health-crisis
15 https://www.nytimes.com/2022/03/19/us/politics/military-mental-health.html
16 https://www.cnn.com/2019/06/06/politics/us-military-deadly-accidents/index.html
17 https://dcas.dmdc.osd.mil/dcas/app/summaryData/deaths/byYearManner
18 https://www.rand.org/content/dam/rand/pubs/research_reports/RR2400/RR2415/RAND_RR2415.pdf
19 https://books.google.com/books?hl=en&lr=&id=CA_JEAAAQBAJ&oi=fnd&pg=PT7&dq=student+centered+learning&ots=mUnEdxj0Nq&sig=erOZO5DUU4_AgmMPP2WGPdBzD9s#v=onepage&q=student%20centered%20learning&f=false
20 https://files.eric.ed.gov/fulltext/ED567875.pdf
21 https://www.airforce.com/training/technical-training

22 CMSgt Morrison, while in AETC, refined this method to determine which competencies an AETC-based scenario required and, consequently, how well an AI chatbot can fulfill that competency need. He is now at the CDAO and works closely with Task Force Lima.
23 https://arxiv.org/abs/2307.10928
24 https://cset.georgetown.edu/publication/ai-education-in-china-and-the-united-states/
25 https://sghexport.shobserver.com/html/baijiahao/2023/12/05/1196748.html . The latter has been particularly useful for English and other foreign language instruction.
26 https://teaching.cornell.edu/sites/default/files/2023-08/Cornell-GenerativeAIForEducation-Report_2.pdf
27 The nation's top-ranked province on the comprehensive development index of basic education informatization, http://jyt.zj.gov.cn/art/2023/7/19/art_1532773_58937425.html
28 http://jyt.zj.gov.cn/art/2023/3/20/art_1532836_58939666.html
29 https://www.globaltimes.cn/page/202303/1287483.shtml
30 http://www.yuexiu.gov.cn/gzjg/qzf/qjyj/jyzl/xw/jyxw/content/post_9270979.html
31 https://www.163.com/dy/article/IMH0G7NH0511PEBT.html
32 http://www.yuexiu.gov.cn/gzjg/qzf/qjyj/jyzl/xw/jyxw/content/post_9270979.html
33 https://www.songshuai.com
34 http://squirrelai.com/#/
35 https://www.technologyreview.com/2019/08/02/131198/china-squirrel-has-started-a-grand-experiment-in-ai-education-it-could-reshape-how-the/
36 https://journals.sagepub.com/doi/full/10.1177/2212585X221121674
37 https://wenku.baidu.com/?_wkts_=1704781136164
38 https://www.whitehouse.gov/briefing-room/statements-releases/2023/05/04/fact-sheet-biden-harris-administration-announces-new-actions-to-promote-responsible-ai-innovation-that-protects-americans-rights-and-safety/
39 China Artificial Intelligence Talent Training Report
40 https://www.airuniversity.af.edu/Portals/10/CASI/documents/Translations/2023-08-07%20ITOW%20Interim%20Measures%20for%20the%20Management%20of%20Generative%20Artificial%20Intelligence%20Services.pdf
41 https://www.airuniversity.af.edu/Portals/10/CASI/documents/Research/Cyber/2023-08-21%20China's%20ChatGPT%20War.pdf
42 http://www.81.cn/yw_208727/16270005.html
43 https://www.nesdnow.navy.mil/docs/NESD%20Amelia%20Fact%20Sheet_v1.1_Dec2022_Final.pdf
44 infotech.com
45 Although we were unable to identify a Chinese military LLM or chatbot platform, an open-source chatbot platform (by Aimoshu) accessing both US and Chinese LLMs appeared during some military-related Baidu searches. Website: chat.aimoshu.cc
46 http://www.81.cn/yw_208727/16270227.html
47 https://myjob.dlmu.edu.cn/search/list/do123/myjob.dlmu.edu.cn/domain/myjob/keyword/ai/page/2
48 https://ai.usajobs.gov/search/results/?cmco=AI&s=relevance&sd=asc&p=1&k=artificial%20intelligence
49 https://jobs.lever.co/dds
50 https://www.dds.mil/taskforcelima
51 https://www.whitehouse.gov/briefing-room/presidential-actions/2023/10/30/executive-order-on-the-safe-secure-and-trustworthy-development-and-use-of-artificial-intelligence/
52 http://www.cppcc.gov.cn/zxww/2023/09/07/ARTI1694054275001277.shtml
53 https://www.gov.cn/xinwen/2019-09/20/content_5431640.htm
54 https://m.163.com/dy/article/I46E93RG0553C0KA.html
55 Robust infrastructure includes labs or companies developing and deploying large-scale generative AI, such as LLMs that can support public use. US allies include the UK, Israel's AI21 laboratory, Canadian startup Cohere, and French startup Mistral. https://m.163.com/dy/article/I46E93RG0553C0KA.html

56 British Army’s Approach to Artificial Intelligence, Oct. 2023
57 https://hai.stanford.edu/news/china-and-united-states-unlikely-partners-ai
58 https://www.guancha.cn/internation/2024_01_12_722048.shtml